

Asynchronous Buck Converter

Design Document — MPPT Charge Controller Power Stage

1. Purpose

This board is the power stage of a solar MPPT (maximum power point tracking) charge controller. Its job is to sit between a solar panel and a load and act as an electronically adjustable step-down (buck) converter. By varying the PWM duty cycle applied to the switch, the converter changes the effective resistance the panel sees, which lets firmware move the panel to its maximum-power operating point regardless of what the load actually wants.

It is asynchronous: the low-side switch of a normal buck is replaced by a Schottky diode. This removes the need for a half-bridge gate-driver IC and a second MOSFET, at the cost of a few percent efficiency lost in the diode. That trade was chosen deliberately to keep the parts count and the breadboard wiring simple.

2. What it should do

- **Step down** a panel voltage (nominal 18 V, up to ~22 V open-circuit) to a lower output set by duty cycle, following $V_{out} \approx D \times V_{in}$.
- **Deliver current to a load** — a 10 Ω power resistor for bench testing — while two INA226 meters read power in and power out.
- **Be steerable by an ESP32**: a single PWM line sets the duty; changing duty is how MPPT is performed.
- **Fail safe at boot**: with the PWM line low or floating, the MOSFET is off. Nothing runs until firmware drives it.

3. How it works

Switch (Q1). A P-channel MOSFET on the high side. Its source sits on the input rail; its drain is the switch node. When the gate is pulled about 10 V below the source, Q1 conducts and connects the input to the inductor.

Gate drive. The ESP32 cannot swing the gate the way a P-channel needs, so an NPN transistor (Q2) level-shifts it. GPIO high turns Q2 on, which pulls the gate down through R2 and switches Q1 on. Because two inversions cancel (GPIO-to-gate, then gate-to-switch-node), firmware duty maps directly to converter duty with no inversion. R3 pulls the gate back up to the source to switch Q1 off; the two zeners clamp gate-to-source to about 10 V so the ± 20 V gate rating is never exceeded.

Freewheel diode (D1). When Q1 turns off, the inductor current must keep flowing. The Schottky gives it a path from ground up to the switch node. A Schottky is used (not an ordinary silicon rectifier) because it switches fast enough at the converter's frequency and drops less voltage.

Inductor and capacitors. L1 smooths current; C_{in}/C_{out} hold the rails steady while the switch chops. Each electrolytic is paired with a 100 nF ceramic to handle the fast switching edges the large cap is too slow for.

Measurement. INA226 #1 sits in series on the input, #2 in series on the output. Comparing the two gives converter efficiency; the input reading is what the MPPT algorithm maximizes.

4. Bill of materials

Ref	Part	Value / spec	Role
Q1	IRF9540	P-channel MOSFET, 100 V	High-side switch
Q2	BC337	NPN transistor	Gate level-shifter
D1	SR560	5 A 60 V Schottky	Freewheel diode
L1	Toroid	330 uH, 3 A sat	Energy storage / smoothing
Cin	Electrolytic	220 uF, 50 V	Input bulk
Cout	Electrolytic	1000 uF, 16 V	Output bulk
C2, C4	Ceramic	100 nF	High-frequency bypass
Z1, Z2	1N4733A	5.1 V zener ×2	Gate clamp (~10 V)
R1	Resistor	1 kΩ	Q2 base
R2	Resistor	220 Ω	Gate pull-down
R3	Resistor	680 Ω	Gate pull-up
—	INA226 ×2	0.02 Ω shunt	Power in / out
—	ESP32	—	PWM controller

5. Key operating figures (10 Ω load)

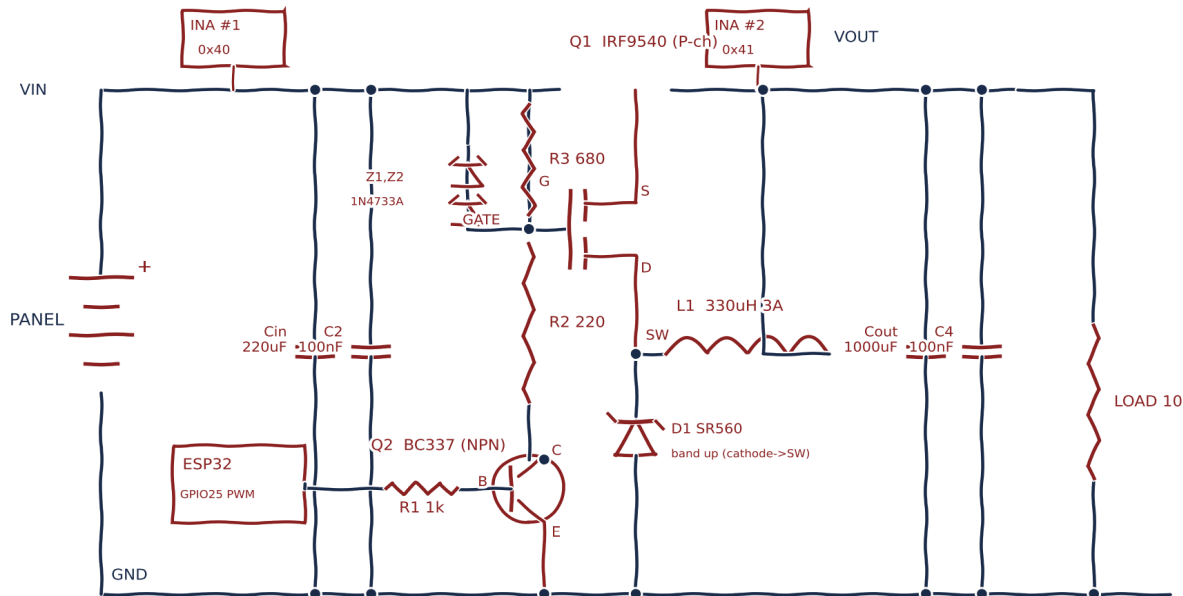
Quantity	On the panel (MPP)	On 19 V bench (test)
Input voltage	~18 V	19 V
Duty at operating point	~0.55	varies, cap at 0.70
Output voltage	~9 V	~12 V at D=0.70
Output current	~0.9 A	~1.2 A at D=0.70
Switching frequency	10 kHz	10 kHz
Efficiency (approx.)	~84 %	~86 %

Hard limit: Cout is rated 16 V. On the 19 V bench supply, keep duty at or below 0.75 (Vout ≈ 14 V) so the output cap is never overvolted. Firmware enforces a duty ceiling. The panel never reaches this because its MPP sits near duty 0.55.

6. Schematic (hand sketch)

Red = components, dark blue = wires and nodes. The single switch symbol of a textbook buck is expanded here into Q1 plus its full gate-drive network (Q2, R1–R3, Z1–Z2). The two INA226 meters tap into the input and output rails in series.

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PWM path: GPIO25 -> R1 -> Q2 base. Q2 ON pulls gate LOW -> P-FET ON. No firmware inversion.

Both INA226 on shared I2C (SDA=21, SCL=22, 3V3, GND). Shunts 0.02 ohm. ESP32 GND ties to circuit GND.

Orientation reminders for the build: D1 band (cathode) points up toward the switch node. Both zener bands point up toward the input rail. Q1 legs are Gate–Drain–Source with the metal tab (internally the drain) kept clear of the power rails. Q2 is collector–base–emitter.